

Proposed Solution Report

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Emotion Detection and Learning Support Engine is an AI-powered application that analyzes a student's text input to identify emotions using Natural Language Processing and a BiLSTM deep learning model. Based on the detected emotion, the system provides personalized learning guidance, stores prediction logs, and visualizes emotion trends through an analytics dashboard. The solution aims to improve student engagement, motivation, and learning outcomes by making educational support emotion-aware.

Project Overview

Objective	To develop an AI-powered application that detects emotions from user text and provides personalized learning guidance using Natural Language Processing and a BiLSTM deep learning model.
Scope	Detect emotions; Classify Joy, Anger, Fear, Sadness, Love, Surprise; Provide personalized guidance; Maintain prediction logs; Display analytics; Streamlit interface.
Problem Statement	Description: Students experience stress/anxiety while learning; existing platforms ignore emotional states. Impact: Reduced efficiency, low motivation, academic stress, lack of support.
Proposed Solution	Approach: Use NLP and BiLSTM to predict emotions and provide guidance via Streamlit. Features: AI detection, NLP, BiLSTM, guidance, confidence scores, logging, analytics.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5, 8 GB RAM, 256 GB SSD
Hardware	Memory (RAM)	Minimum 8 GB RAM
Hardware	Storage	Minimum 2 GB free disk space
Software	Programming Language	Python 3.11
Software	Frameworks	Streamlit, TensorFlow, Keras
Software	Libraries	NumPy, Pandas, Scikit-learn, Matplotlib, Pickle
Software	Development Environment	Visual Studio Code, Jupyter Notebook
Software	Version Control	Git and GitHub
Data	Data Source	Emotion text dataset (CSV)
Data	Dataset Format	CSV (Text, Emotion Label)

Data	Model Files	BiLSTM model (.h5), Tokenizer (.pkl), Label Encoder (.pkl)
------	-------------	--